

Example:

What constitute a point x ?

What are the constraints function $g_i(x)$? Under subject to...

$\mathcal{E}$ is the equality, $\mathcal{I}$ is the inequality set

$(0, 125, 50, 50)$ feasible or infeasible?

2 plug in each x into constraint to see if its possible.

Active or inactive, inactive if not equal but still

↳ on a boundary as defined by a solution space

$$g_1(x) = x_1 + 2x_2 + 3x_3 - 6 = 0$$

What is $\mathcal{E}$ and the set $\mathcal{I}$

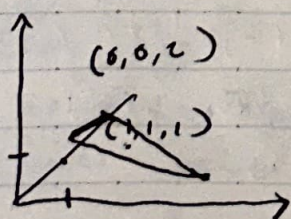
		Feasible?	Active?	Boundary?	Ans
i	$(0, 0, 2)^T$	✓	✓	✓	0
ii	$(3, 0, 1)^T$	✓	✓	✓	0
iii	$(0, 2, 1)^T$	X	X	X	1
iv	$(1, 1, 1)^T$	✓	✓	✓	0

Set S

~~feasible~~ feasible points [

Boundary $[(0, 0, 2), (3, 0, 1), (4, 1, 0)]$

Definitions to declare



- Feasible

- Active

- Boundary

What is the differences between active & boundary?

Optimization by moving in feasible directions

- General Algorithm 1

1. initial guess
2. for $k=0, 1, \dots$
 - o if x_k is optimal, stop
 - o if not move to x_{k+1}

- General Algorithm 2

1. initial guess
2. for $k=0, 1, \dots$
 - o if x_k is optimal, stop
 - o determine search direction
 - o determine a step length
 - $x_{k+1} = x_k + \alpha_k p_k$

Determining Direction p_k

- p_k is referred to descent location (for minimization)

$$f(x_k + \alpha p_k) < f(x_k) \text{ for } 0 < \alpha \leq \epsilon$$

$\nwarrow$ k^{th} step

$\nwarrow$ step length

(small amount $\times$ direction)

$\nwarrow$

Some arbitrary small number

"Whenever we make a small step value is less than where we start"

- Nonlinear is all about making constant improvement steps

- Linear $f(x) = c^T x$ (prog. prob.) (for minimization)

$$c^T x_k + \alpha c^T p_k < c^T x_k \Rightarrow c^T p_k < 0 \text{ feasible set } S$$

$\nwarrow$ only move in that direction if unfeasible,

Feasible-point Method

$$p(x) = x_1 + x_2 = 1 \quad c^T x = b_i \text{ with } c_i = (1, 1)^T \text{ and } b_i = 1$$

if we move in direction p then $c_i^T (x + \alpha p) = b_i$ only if $c_i^T p = 0$ or $p_1 + p_2 = 0$, p is the null space of c_i .

if switch to $x_1 + x_2 \geq$ then $c_i^T p \geq 0$ to stay feasible

- Null SPACE is important for finding feasible points around the boundary.

Feasible-point example

Minimize $2x_1 - x_2$

subject to $x_1 + x_2 \leq 2, x_1, x_2 \geq 0$

NOTE: $f(x) = c^T x$ such that $c = (2, -1)$

Start with guess $x^0 = (1, 1)$ ← always start with active point.

Feasible decent $x^{(0)}$ is an active feasible point so $p^{(0)}$ such that $c^T p^{(0)} < 0$ while also meeting active constraint.

$$\begin{aligned} 2p_1^{(0)} - p_2^{(0)} &< 0 & x_1 + x_2 &\leq 2 \quad \text{so we also need} \\ p_1^{(0)} + p_2^{(0)} &\leq 0 \end{aligned}$$

Try $p^{(0)} = (-1, 1)$; Now $x^{(1)} = x^{(0)} + \alpha p^{(0)} = (1 - \alpha, 1 + \alpha)$

$$x_1^{(1)} + x_2^{(1)} = 2 \leq 2 \quad \text{for any value of } \alpha > 0$$

$$x_1^{(1)} = 1 - \alpha \geq 0 \quad \text{for } \alpha \leq 1$$

$$x_2^{(1)} = 1 + \alpha \geq 0 \quad \alpha > 0$$

So $\alpha = 1$, and then $x^{(1)} = (0, 2)$ which is optimal.

Example 2:

min $2x_1 - x_2$

s.t $x_1 + x_2 \leq 2$

$x_1, x_2 \geq 0$

~~choose~~ $p^{(0)} = (-1, 0)$ initial step direction $(1, 1) = x^0$

$$x' = x^0 + \alpha p^{(0)} = (1 - \alpha, 1) \quad \leftarrow \text{new slope?}$$

$$x_1' + x_2' = 2 - \alpha \leq 2$$

$$x_1' = 1 - \alpha \geq 0 \quad \alpha \leq 1$$

$$x_2' = 1 \geq 0$$

How did we determine $x^{(0)}$ & $x^{(1)}$

results in $x' = (0, 1)$

$$p^{(1)} = (0, 1) \rightarrow x^{(2)} = x^{(1)} + \alpha p^{(1)} = (0, a+1)$$

'here we can move as much as $\alpha = 1$ ', therefore $x^2 = (0, 2)$
magnitude α should be as ~~big~~ as possible
big!

PDF ON Brightspace Who! HW due next Wednesday.